Graphs 4 — Quadratic graphs · Solving $ax^2 + bx + c = 0$ graphically

Squared paper needed. Plot the points carefully and join them with a *smooth curve* — never straight lines. Marks in brackets. ★ = harder question.

1 Copy and complete the table, then draw the graph of $y = x^2 - 2x - 3$ for $-2 \le x \le 4$. [6]

x	-2	-1	0	1	2	3	4
y		0		-4		0	

Use your graph to write down the solutions of $x^2 - 2x - 3 = 0$.

> $x = -2$: $4 + 4 - 3 = 5$; $x = 0$: -3; $x = 2$: $4 - 4 - 3 = -3$; $x = 4$: $16 - 8 - 3 = 5$. Table: $5,\ 0,\ -3,\ -4,\ -3,\ 0,\ 5$. A U-shaped parabola, lowest point $(1, -4)$, symmetric about $x = 1$.
> $x^2 - 2x - 3 = 0$ where the curve crosses the x-axis: **$x = -1$ and $x = 3$**.

2 **a** Copy and complete the table, then draw the graph of $y = 2x^2 + x - 3$ for $-3 \le x \le 2$. [8]

x	-3	-2	-1	0	1	2
$2x^2$		8		0	2	
$+x$		-2		0	1	
-3	-3	-3	-3	-3	-3	-3
y		3		-3	0	

b Use your graph to solve the equation $2x^2 + x - 3 = 0$.

> **a** $2x^2$: $18,\ 8,\ 2,\ 0,\ 2,\ 8$; $+x$: $-3, -2, -1, 0, 1, 2$; y: $12,\ 3,\ -2,\ -3,\ 0,\ 7$.
> **b** The curve crosses the x-axis at **$x = -1.5$ and $x = 1$**. (Check: $2(2.25) - 1.5 - 3 = 0$ ✓)

3 The profit p (€ millions) of a start-up company t years after launch is modelled by $p = 5t - kt^2$ for $0 \le t \le 10$, where k is a constant. [11]

t	0	2	4	6	8	10
p	0		12			0

a Use the table to show that $k = 0.5$, then copy and complete the table.

b Draw the graph of p against t.

c Use the graph to estimate **(i)** the greatest profit and when it occurs, **(ii)** the period during which the profit is at least €8 million.

a At $t = 4$: $20 - 16k = 12 \Rightarrow 16k = 8 \Rightarrow k = 0.5$. (Check $t = 10$: $50 - 50 = 0$ ✓.) Then $p = 5t - 0.5t^2$: $t = 2 \to 8$, $t = 6 \to 12$, $t = 8 \to 8$.

b An inverted parabola (∩) from $(0, 0)$ up to its peak and back to $(10, 0)$, symmetric about $t = 5$.

c (i) Greatest profit at the top of the curve, $t = 5$: Math input error million. **(ii)** $p \ge 8$ for $2 \le t \le 8$ — from the 2nd to the 8th year.

Total: 25 marks

Check: a quadratic graph is a parabola — U if $a > 0$, ∩ if $a < 0$ — and symmetric · the solutions of $ax^2 + bx + c = 0$ are where the curve crosses the x-axis (two, one or none) · read the greatest/least value at the turning point.